10/05/2025

1. Description

1.1. Project

Project Name	CBXTouchGFx
Board Name	custom
Generated with:	STM32CubeMX 6.14.1
Date	05/10/2025

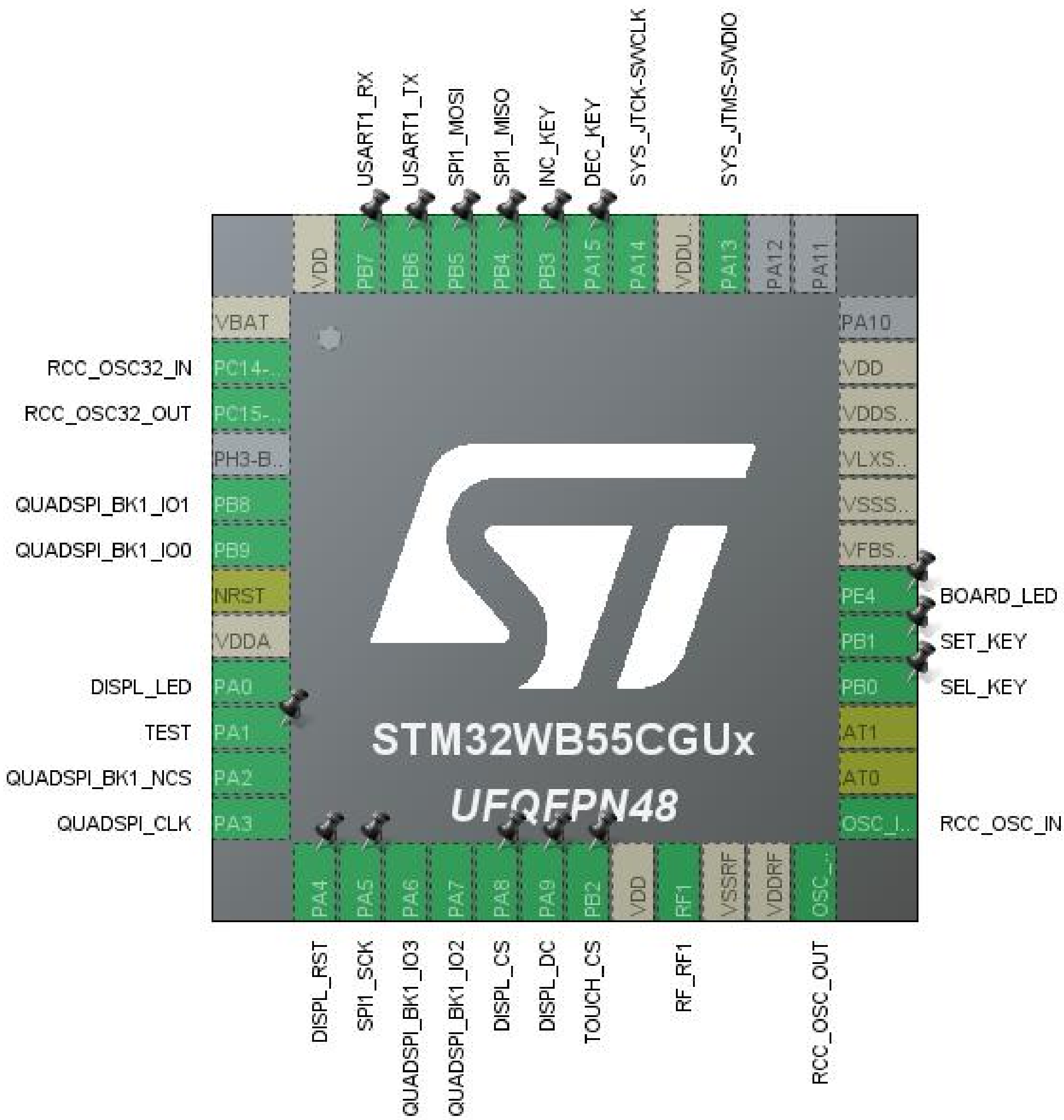
1.2. MCU

MCU Series	STM32WB
MCU Line	STM32WBx5
MCU name	STM32WB55CGUx
MCU Package	UFQFPN48
MCU Pin number	48

1.3. Core(s) information

Core(s)	ARM Cortex-M4

2. Pinout Configuration



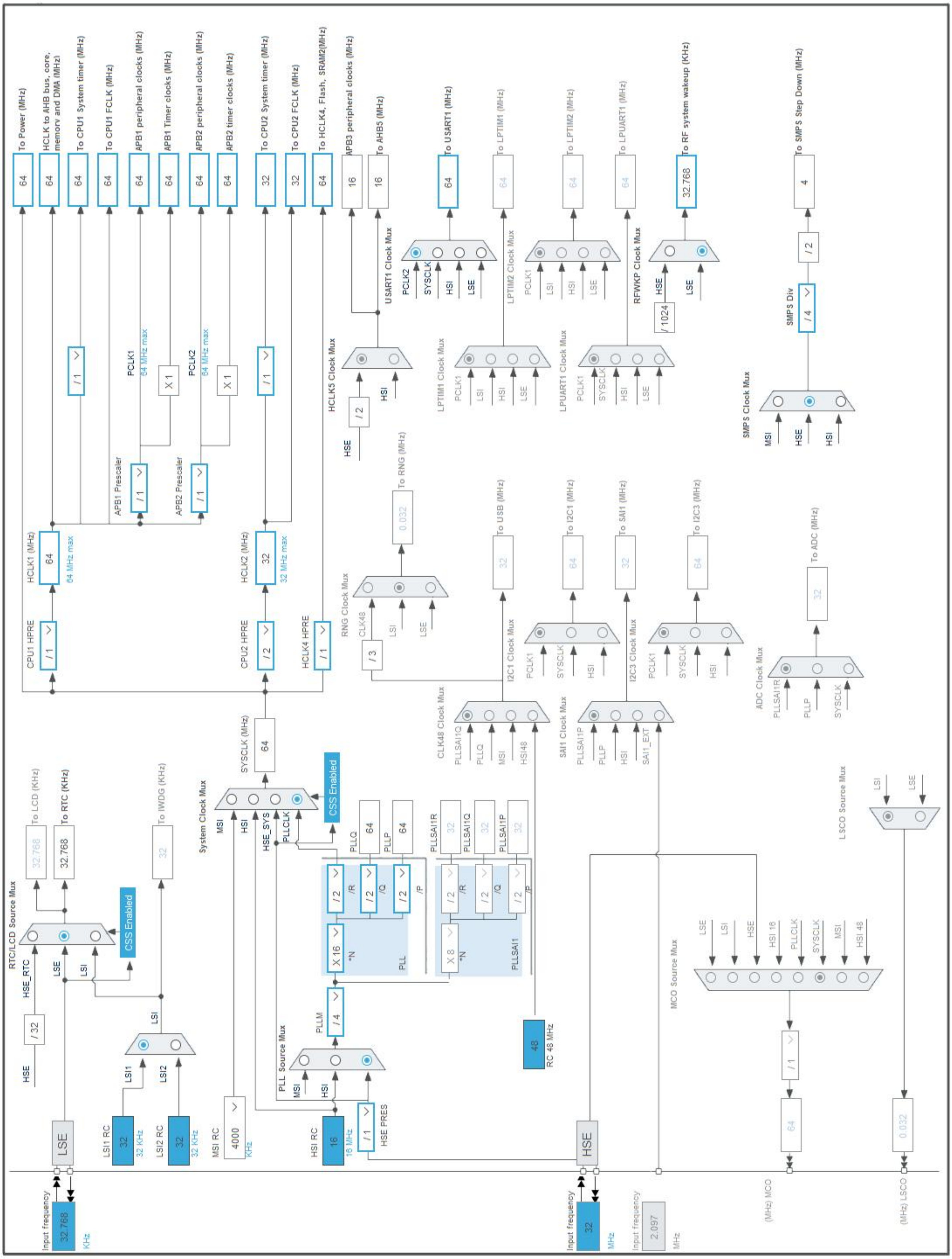
3. Pins Configuration

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
2	PC14-OSC32_IN	I/O	RCC_OSC32_IN	
3	PC15-OSC32_OUT	I/O	RCC_OSC32_OUT	
5	PB8	I/O	QUADSPI_BK1_IO1	
6	PB9	I/O	QUADSPI_BK1_IO0	
7	NRST	Reset		
8	VDDA	Power		
9	PA0	I/O	TIM2_CH1	DISPL_LED
10	PA1 *	I/O	GPIO_Output	TEST
11	PA2	I/O	QUADSPI_BK1_NCS	
12	PA3	I/O	QUADSPI_CLK	
13	PA4 *	I/O	GPIO_Output	DISPL_RST
14	PA5	I/O	SPI1_SCK	
15	PA6	I/O	QUADSPI_BK1_IO3	
16	PA7	I/O	QUADSPI_BK1_IO2	
17	PA8 *	I/O	GPIO_Output	DISPL_CS
18	PA9 *	I/O	GPIO_Output	DISPL_DC
19	PB2 *	I/O	GPIO_Output	TOUCH_CS
20	VDD	Power		
21	RF1	MonoIO	RF_RF1	
22	VSSRF	Power		
23	VDDRF	Power		
24	OSC_OUT	MonoIO	RCC_OSC_OUT	
25	OSC_IN	MonoIO	RCC_OSC_IN	
26	AT0	NC		
27	AT1	NC		
28	PB0	I/O	GPIO_EXTI0	SEL_KEY
29	PB1	I/O	GPIO_EXTI1	SET_KEY
30	PE4 *	I/O	GPIO_Output	BOARD_LED
31	VFBSMPS	Power		
32	VSSSMPS	Power		
33	VLXSMPS	Power		
34	VDDSMPS	Power		
35	VDD	Power		
39	PA13	I/O	SYS_JTMS-SWDIO	
40	VDDUSB	Power		

Pin Number UFQFPN48	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
41	PA14	I/O	SYS_JTCK-SWCLK	
42	PA15	I/O	GPIO_EXTI15	DEC_KEY
43	PB3	I/O	GPIO_EXTI3	INC_KEY
44	PB4	I/O	SPI1_MISO	
45	PB5	I/O	SPI1_MOSI	
46	PB6	I/O	USART1_TX	
47	PB7	I/O	USART1_RX	
48	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32WB
Line	STM32WBx5
MCU	STM32WB55CGUx
Datasheet	DS11929_Rev3

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

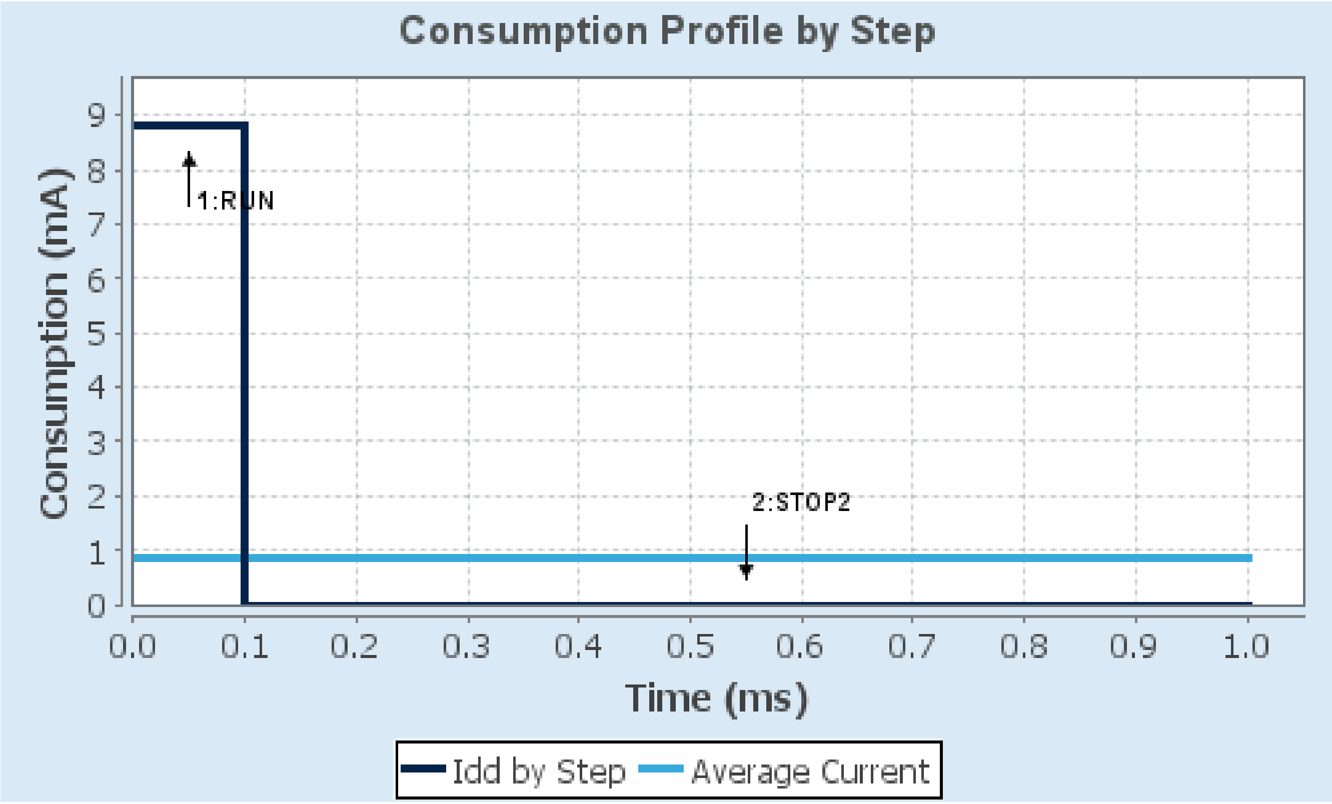
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP2
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	Range1-High	NoRange
Fetch Type	SRAM1/Flash-PowerDown	FLASH/ART/CACHE
CPU Frequency	64 MHz	0 Hz
Clock Configuration	HSI PLL Regulator_ON	ALL CLOCKS OFF Regulator_ON
Clock Source Frequency	16 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	8.8 mA	1.85 µA
Duration	0.1 ms	0.9 ms
DMIPS	80.0	0.0
Ta Max	104.34	105
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	881.66 µA
Battery Life	5 months, 7 days, 21 hours	Average DMIPS	8.0 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
9
8
7
6
5
4
3
2
1
0
1:RUN
2:STOP2
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	CBXTouchGFx
Project Folder	D:\ARM\Learning\CBXTouchGFx
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_WB V1.22.1
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x2000
Minimum Stack Size	0x4000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_SPI1_Init	SPI1
5	MX_CRC_Init	CRC
6	MX_TIM2_Init	TIM2
7	MX_QUADSPI_Init	QUADSPI
8	MX_USART1_UART_Init	USART1
9	MX_IPCC_Init	IPCC
10	MX_RTC_Init	RTC
11	APPE_Init	STM32_WPAN

Rank	Function Name	Peripheral Instance Name
14	MX_TouchGFX_Init	STMicroelectronics.X-CUBE-TOUCHGFX.4.25.0
15	MX_TouchGFX_Process	STMicroelectronics.X-CUBE-TOUCHGFX.4.25.0
16	MX_RF_Init	RF

3. Peripherals and Middlewares Configuration

3.1. CRC

mode: Activated

3.1.1. Parameter Settings:

Basic Parameters:	
Default Polynomial State	Enable
Default Init Value State	Enable
Advanced Parameters:	
Input Data Inversion Mode	None
Output Data Inversion Mode	Disable
Input Data Format	Bytes

3.2. HSEM

mode: Activated

3.3. IPCC

mode: Activated

3.4. MEMORYMAP

mode: Activated

3.5. QUADSPI

QuadSPI Mode: Bank1 with Quad SPI Lines

3.5.1. Parameter Settings:

General Parameters:	
Clock Prescaler	**1 ***
Fifo Threshold	**4 ***
Sample Shifting	**Sample Shifting Half Cycle ***
Flash Size	**23 ***
Chip Select High Time	1 Cycle
Clock Mode	Low
Flash ID	Flash ID 1
Dual Flash	Disabled

3.6. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

3.6.1. Parameter Settings:

System Parameters:	
VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	3 WS (4 CPU cycle)
RCC Parameters:	
HSI Calibration Value	16
MSI Calibration Value	0
MSI Auto Calibration	Disabled
MSI State	Enabled
HSI State	Enabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
LSE Drive Capability	LSE oscillator medium high drive capability
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
Peripherals Clock Configuration:	
Generate the peripherals clock configuration	TRUE

3.7. RF

mode: Activate RF1

3.8. RTC

mode: Activate Clock Source

mode: Activate Calendar

WakeUp: Internal WakeUp

3.8.1. Parameter Settings:

General:

Hour Format	Hourformat 24
Asynchronous Predivider value	CFG_RTC_ASYNCH_PRESCALER
Synchronous Predivider value	CFG_RTC_SYNCH_PRESCALER

Calendar Time:

Data Format	**Binary data format ***
Hours	0
Minutes	0
Seconds	0
Day Light Saving: value of hour adjustment	Daylightsaving None
Store Operation	Storeoperation Reset

Calendar Date:

Week Day	Monday
Month	January
Date	1
Year	0

Wake UP:

Wake Up Clock	RTCCLK / 16
Wake Up Counter	0

3.9. SEQUENCER
mode: Enabled

3.10. SPI1
Mode: Full-Duplex Master
3.10.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	**8 Bits ***
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	**32.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSSP Mode	**Disabled ***

NSS Signal Type	Software

3.11. SYS
Debug: Serial Wire
Timebase Source: SysTick

3.12. TIM2
Clock Source : Internal Clock
Channel1: PWM Generation CH1

3.12.1. Parameter Settings:

Counter Settings:	
Prescaler (PSC - 16 bits value)	**64000-1 ***
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	**250 ***
Internal Clock Division (CKD)	**Division by 4 ***
auto-reload preload	**Enable ***
Trigger Output (TRGO) Parameters:	
Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	**Enable (CNT_EN) ***
Clear Input:	
Clear Input Source	Disable
PWM Generation Channel 1:	
Mode	PWM mode 1
Pulse (32 bits value)	**50 ***
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High

3.13. TINY_LPM
mode: Enabled

3.14. USART1
Mode: Asynchronous

3.14.1. Parameter Settings:

Basic Parameters:

Baud Rate	**38400 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable
TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.15. STM32_WPAN
mode: BLE
3.15.1. BLE Applications and Services:

BLE Wireless Stack:

BLE Wireless Stack	Full

BLE Application Type:

BLE Application Type	Server profile

Server Mode:

BT SIG Beacon	Disabled
BT SIG Blood Pressure Sensor	Disabled
BT SIG Health Thermometer Sensor	Disabled
BT SIG Heart Rate Sensor	Disabled
Custom P2P Server	**Disabled ***

Custom Template	**Enabled ***

BLE Services Configuration:

The device needs to support the Peripheral Role	1
The device needs to support the Central Role	0
BLE_CFG_SVC_MAX_NBR_CB	7
BLE_CFG_CLT_MAX_NBR_CB	0

3.15.2. Configuration:

HW Timer Server:

CFG_HW_TS_MAX_NBR_CONCURRENT_TIMER	6
CFG_HW_TS_NVIC_RTC_WAKEUP_IT_PREEMPTPRIO	3
CFG_HW_TS_NVIC_RTC_WAKEUP_IT_SUBPRIO	0
CFG_HW_TS_USE_PRIMASK_AS_CRITICAL_SECTION	1
CFG_HW_TS_RTC_HANDLER_MAX_DELAY	(10 * (LSI_VALUE/1000))
CFG_HW_TS_RTC_WAKEUP_HANDLER_ID	RTC_WKUP_IRQn

HW UART:

CFG_HW_LPUART1_ENABLED	Disabled
CFG_HW_LPUART1_DMA_TX_SUPPORTED	Disabled
CFG_HW_USART1_ENABLED	Disabled
CFG_HW_USART1_DMA_TX_SUPPORTED	Disabled

Generic parameters:

CFG_HW_RESET_BY_FW	Disabled
CFG_USE_SMPS	Disabled
CFG_LPM_SUPPORTED	Disabled
CFG_DEBUGGER_SUPPORTED	Enabled
CFG_DEBUG_BLE_TRACE	Disabled
CFG_DEBUG_APP_TRACE	Disabled
CFG_DEBUG_TRACE_LIGHT	Disabled
CFG_DEBUG_TRACE_FULL	Disabled
DBG_TRACE_USE_CIRCULAR_QUEUE	Enabled
DBG_TRACE_MSG_QUEUE_SIZE	4096
MAX_DBG_TRACE_MSG_SIZE	1024

Application parameters:

CFG_TX_POWER	-0.15dBm (0x18)
CFG_DEBUG_TRACE_UART	You need to activate either CFG_HW_UART1 or CFG_HW_LPUART1(when available)
CFG_CONSOLE_MENU	You need to activate either CFG_HW_UART1 or CFG_HW_LPUART1(when available)
CFG_ADV_BD_ADDRESS	**0x11aabbccddee ***

CFG_FAST_CONN_ADV_INTERVAL_MIN	80
CFG_FAST_CONN_ADV_INTERVAL_MAX	100
CFG_LP_CONN_ADV_INTERVAL_MIN	1000
CFG_LP_CONN_ADV_INTERVAL_MAX	2500
CFG_IO_CAPABILITY	Display Yes No (0x01)
CFG_MITM_PROTECTION	MITM protection required (0x01)
L2CAP_REQUEST_NEW_CONN_PARAM	0
CFG_RTCCLK_DIVIDER_CONF	0
CFG_RTCCLK_DIV	16
CFG_RTC_WUCKSEL_DIVIDER	0
CFG_RTC_ASYNCH_PRESCALER	**0x0F ***
CFG_RTC_SYNCH_PRESCALER	**0x7FFF ***
CFG_BLE_NUM_LINK	2
CFG_BLE_NUM_GATT_SERVICES	8
CFG_BLE_NUM_GATT_ATTRIBUTES	68
CFG_BLE_MAX_ATT_MTU	156
CFG_BLE_ATT_VALUE_ARRAY_SIZE	1344
CFG_BLE_DATA_LENGTH_EXTENSION	Enabled
CFG_BLE_PERIPHERAL_SCA	500
CFG_BLE_CENTRAL_SCA	0
CFG_BLE_HSE_STARTUP_TIME	**0x148 ***
CFG_BLE_MAX_CONN_EVENT_LENGTH	**0xFFFFFFFF ***
CFG_BLE_VITERBI_MODE	Enabled
CFG_BLE_OPTIONS	BLE stack Options flags:
- CFG_BLE_OPTIONS_LL	SHCI_C2_BLE_INIT_OPTIONS_LL_HOST
- CFG_BLE_OPTIONS_SVC	SHCI_C2_BLE_INIT_OPTIONS_WITH_SVC_CHANGE_DESC
- CFG_BLE_OPTIONS_DEVICE_NAME	SHCI_C2_BLE_INIT_OPTIONS_DEVICE_NAME_RW
- CFG_BLE_OPTIONS_EXT_ADV	SHCI_C2_BLE_INIT_OPTIONS_NO_EXT_ADV
- CFG_BLE_OPTIONS_CS_ALGO	SHCI_C2_BLE_INIT_OPTIONS_NO_CS_ALGO2
- CFG_BLE_OPTIONS_GATTDB_NVM	SHCI_C2_BLE_INIT_OPTIONS_FULL_GATTDB_NVM
- CFG_BLE_OPTIONS_GATT_CACHING	SHCI_C2_BLE_INIT_OPTIONS_GATT_CACHING_NOTUSED
- CFG_BLE_OPTIONS_POWER_CLASS	SHCI_C2_BLE_INIT_OPTIONS_POWER_CLASS_2_3
- CFG_BLE_OPTIONS_APPEARANCE	SHCI_C2_BLE_INIT_OPTIONS_APPEARANCE_READONLY
- CFG_BLE_OPTIONS_ENHANCED_ATT	SHCI_C2_BLE_INIT_OPTIONS_ENHANCED_ATT_NOTSUPPORTED

CFG_BLE_MAX_COC_INITIATOR_NBR	32
CFG_BLE_MIN_TX_POWER	0
CFG_BLE_MAX_TX_POWER	0
CFG_BLE_RX_MODEL_CONFIG	SHCI_C2_BLE_INIT_RX_MODEL_AGC_RSSI_LEGACY
CFG_BLE_MAX_ADV_SET_NBR	3
CFG_BLE_MAX_ADV_DATA_LEN	1650
CFG_BLE_TX_PATH_COMPENS	0
CFG_BLE_RX_PATH_COMPENS	0
CFG_BLE_CORE_VERSION	SHCI_C2_BLE_INIT_BLE_CORE_5_4
CFG_TLBLE_EVT_QUEUE_LENGTH	5
CFG_TLBLE_MOST_EVENT_PAYLOAD_SIZE	255
Debug options:	
BLE_DBG_APP_EN	Disabled

3.15.3. BLE Advertising:

Advertising configuration:	
Advertising Type	Undirected scannable and connectable(0x00)
CFG_IDENTITY_ADDRESS	GAP_PUBLIC_ADDR
CFG_PRIVACY	Disabled
Advertising Filter	No white list(0x00)
Peripheral: Advertise and connectable	**Yes (0x01) ***
Broadcaster: Advertise and non-connectable	No (0x00)
Central: Scan and connect	No (0x00)
Observer: Scan	No (0x00)
CFG_GAP_DEVICE_NAME	**CBX_1 ***
CFG_GAP_DEVICE_NAME_LENGTH	**5 ***
Advertising elements:	
ad_data[] length	22
Include AD_TYPE_TX_POWER_LEVEL element	**Yes ***
AD_TYPE_TX_POWER_LEVEL_LENGTH	2
AD_TYPE_TX_POWER_LEVEL	(0x18) /* -0.15dBm */
Include AD_TYPE_COMPLETE_LOCAL_NAME element	**Yes ***
AD_TYPE_COMPLETE_LOCAL_NAME_LENGTH	9
AD_TYPE_COMPLETE_LOCAL_NAME	XX-STM32
Include AD_TYPE_SHORTENED_LOCAL_NAME element	No
Include AD_TYPE_APPEARANCE element	No
Include AD_TYPE_ADVERTISING_INTERVAL element	No
Include AD_TYPE_LE_ROLE element	No

Include AD_TYPE_16_BIT_SERV_UUID_CMPLT_LIST element	**Yes ***
AD_TYPE_16_BIT_SERV_UUID_CMPLT_LIST_LENGTH	3
ServiceClass UUIDs number	1
UUID 1	**FE 00 ***
Include AD_TYPE_128_BIT_SERV_UUID_CMPLT_LIST element	No
Include AD_TYPE_SLAVE_CONN_INTERVAL element	No
Include AD_TYPE_URI element	No
Include AD_TYPE_MANUFACTURER_SPECIFIC_DATA element	**Yes ***
AD_TYPE_MANUFACTURER_SPECIFIC_DATA_LENGTH	4
Company identifier	30,00
Number of user defined data item(s)	1
User defined data 1	00
Comment data 1	

3.15.4. BLE Pairing:

Pairing parameters:

PAIRING_PARAMETERS	**ON ***
CFG_BONDING_MODE	No-bonding mode(0x00)
CFG_USED_FIXED_PIN	Use a fixed pin (0x00)
CFG_FIXED_PIN	111111
CFG_ENCRYPTION_KEY_SIZE_MAX	16
CFG_ENCRYPTION_KEY_SIZE_MIN	8
CFG_SC_SUPPORT	Secure Connections Paring supported but optional (0x01)
CFG_BLE_IR	12, 34, 56, 78, 9A, BC, DE, F0, 12, 34, 56, 78, 9A, BC, DE, F0
CFG_BLE_ER	FE, DC, BA, 09, 87, 65, 43, 21, FE, DC, BA, 09, 87, 65, 43, 21
CFG_KEYPRESS_NOTIFICATION_SUPPORT	Keypress notification not supported (0x00)

3.15.5. BLE GATT:

Services configuration:

Number of services	**1 ***

Service1:

Service long name	**My_CBX_Server ***
Service short name	**mycbx ***

3.15.6. Service1:

Service1:

Number of characteristics	**2 ***
UUID type	128 bits UUID(0x02)
UUID 128 input type	reduced
UUID	**FE 01 ***
Type	Primary Service(0x01)
Service max attributes record(s)	7

Characteristic1 general:

Characteristic long name	**GPS_DATA ***
Characteristic short name	**my_gps ***
UUID type	128 bits UUID(0x02)
UUID 128 input type	reduced
UUID	**FE 41 ***
Value length	**84 ***
Length characteristic	**Variable ***
Encryption Key Size	**0x10 ***
Update char value offset	0

Characteristic1 properties:

CHAR_PROP_BROADCAST	No
CHAR_PROP_READ	No
CHAR_PROP_WRITE_WITHOUT_RESP	No
CHAR_PROP_WRITE	No
CHAR_PROP_NOTIFY	**Yes ***
CHAR_PROP_INDICATE	No

Characteristic1 permissions:

ATTR_PERMISSION_AUTHEN_READ	No
ATTR_PERMISSION_AUTHOR_READ	No
ATTR_PERMISSION_ENCRY_READ	No
ATTR_PERMISSION_AUTHEN_WRITE	No
ATTR_PERMISSION_AUTHOR_WRITE	No
ATTR_PERMISSION_ENCRY_WRITE	No

Characteristic1 GATT events:

GATT_NOTIFY_ATTRIBUTE_WRITE	**No ***
GATT_NOTIFY_WRITE_REQ_AND_WAIT_FOR_APPL_RESP	**No ***
GATT_NOTIFY_READ_REQ_AND_WAIT_FOR_APPL_RESP	**No ***
GATT_NOTIFY_NOTIFICATION_COMPLETION	**Yes ***

Characteristic2 general:

Characteristic long name	**LAT_DATA ***
Characteristic short name	**la_dat ***
UUID type	128 bits UUID(0x02)
UUID 128 input type	reduced
UUID	**FE 42 ***
Value length	**9 ***
Length characteristic	Constant
Encryption Key Size	**0x10 ***
Update char value offset	0

Characteristic2 properties:

CHAR_PROP_BROADCAST	No
CHAR_PROP_READ	No
CHAR_PROP_WRITE_WITHOUT_RESP	No
CHAR_PROP_WRITE	No
CHAR_PROP_NOTIFY	**Yes ***
CHAR_PROP_INDICATE	No

Characteristic2 permissions:

ATTR_PERMISSION_AUTHEN_READ	No
ATTR_PERMISSION_AUTHOR_READ	No
ATTR_PERMISSION_ENCRY_READ	No
ATTR_PERMISSION_AUTHEN_WRITE	No
ATTR_PERMISSION_AUTHOR_WRITE	No
ATTR_PERMISSION_ENCRY_WRITE	No

Characteristic2 GATT events:

GATT_NOTIFY_ATTRIBUTE_WRITE	**No ***
GATT_NOTIFY_WRITE_REQ_AND_WAIT_FOR_APPL_RESP	**No ***
GATT_NOTIFY_READ_REQ_AND_WAIT_FOR_APPL_RESP	**No ***
GATT_NOTIFY_NOTIFICATION_COMPLETION	**Yes ***

3.16. STMicroelectronics.X-CUBE-TOUCHGFX.4.25.0 mode: GraphicsJjApplication

3.16.1. TouchGFX Generator:

Display:

Interface	Custom
Framebuffer Pixel Format	RGB565
Width	**320 ***

Height	**240 ***
Framebuffer Strategy	**Partial Buffer - GRAM display ***
Number of Blocks	3
Block Size	2048

Driver:

Application Tick Source	Custom
Use DMA2D Accelerator (ChromART)	No
Real-Time Operating System	No OS

Additional Features:

Partial Framebuffer VSync	Disabled
External Data Reader	Disabled
Vector Rendering	Disabled

Video Decoding:

Type	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
QUADSPI	PB8	QUADSPI_BK1_IO1	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB9	QUADSPI_BK1_IO0	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PA2	QUADSPI_BK1_NCS	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PA3	QUADSPI_CLK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PA6	QUADSPI_BK1_IO3	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PA7	QUADSPI_BK1_IO2	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
RCC	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
	OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
RF	RF1	RF_RF1	n/a	n/a	n/a	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB4	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
TIM2	PA0	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	DISPL_LED
USART1	PB6	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB7	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
GPIO	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TEST
	PA4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DISPL_RST

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PA8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DISPL_CS
	PA9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	DISPL_DC
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	TOUCH_CS
	PB0	GPIO_EXTI0	**External Interrupt Mode with Falling edge trigger detection**	**Pull-up ***	n/a	SEL_KEY
	PB1	GPIO_EXTI1	**External Interrupt Mode with Falling edge trigger detection**	**Pull-up ***	n/a	SET_KEY
	PE4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BOARD_LED
	PA15	GPIO_EXTI15	**External Interrupt Mode with Falling edge trigger detection**	**Pull-up ***	n/a	DEC_KEY
	PB3	GPIO_EXTI3	**External Interrupt Mode with Falling edge trigger detection**	**Pull-up ***	n/a	INC_KEY

4.2. DMA configuration

DMA request	Stream	Direction	Priority
SPI1_TX	DMA1_Channel1	Memory To Peripheral	Low
USART1_RX	DMA1_Channel2	Peripheral To Memory	Low

SPI1_TX: DMA1_Channel1 DMA request Settings:

Mode:	Normal
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

USART1_RX: DMA1_Channel2 DMA request Settings:

Mode:	**Circular ***
Peripheral Increment:	Disable
Memory Increment:	**Enable ***
Peripheral Data Width:	Byte
Memory Data Width:	Byte

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Prefetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
RTC wake-up interrupt through EXTI line 19	true	0	0
EXTI line0 interrupt	true	0	0
EXTI line1 interrupt	true	0	0
EXTI line3 interrupt	true	0	0
DMA1 channel1 global interrupt	true	0	0
DMA1 channel2 global interrupt	true	0	0
SPI1 global interrupt	true	0	0
USART1 global interrupt	true	0	0
EXTI line[15:10] interrupts	true	0	0
IPCC RX occupied interrupt	true	0	0
IPCC TX free interrupt	true	0	0
HSEM global interrupt	true	0	0
PVD/PVM0/PVM2 interrupts through EXTI lines 16/31/33	unused		
RTC tamper and time stamp, CSS on LSE interrupts through EXTI line 18	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
CPU2 SEV interrupt through EXTI line 40 and PWR CPU2 HOLD wake-up interrupt	unused		
TIM2 global interrupt	unused		
PWR switching on the fly, end of BLE activity, end of 802.15.4 activity, end of critical radio phase interrupt	unused		
QUADSPI global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init	Generate IRQ	Call HAL handler

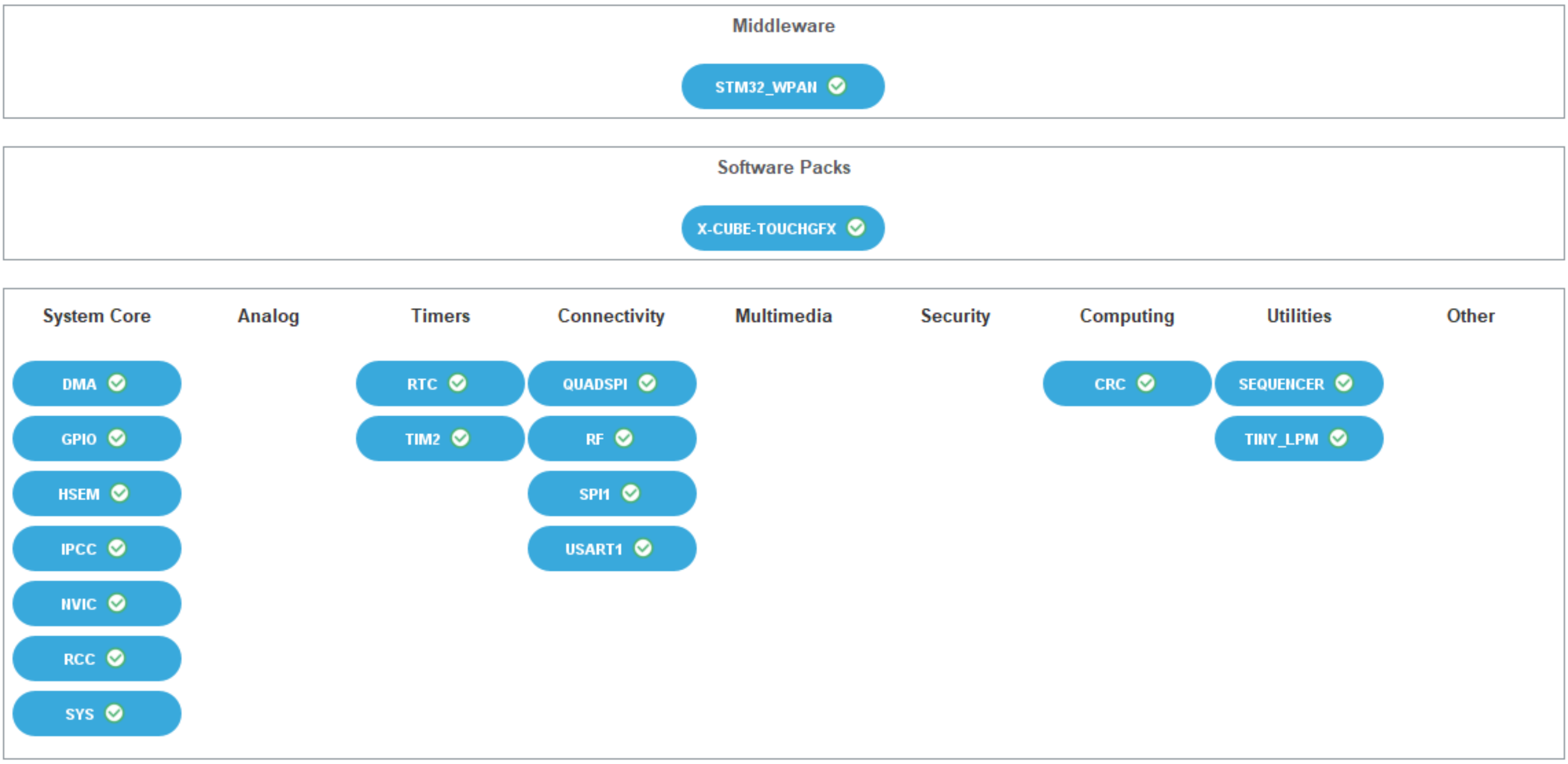
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	true
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Prefetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
RTC wake-up interrupt through EXTI line 19	false	true	true
EXTI line0 interrupt	false	true	true
EXTI line1 interrupt	false	true	true
EXTI line3 interrupt	false	true	true
DMA1 channel1 global interrupt	false	true	true
DMA1 channel2 global interrupt	false	true	true
SPI1 global interrupt	false	true	true
USART1 global interrupt	false	true	true
EXTI line[15:10] interrupts	false	true	true
IPCC RX occupied interrupt	false	true	true
IPCC TX free interrupt	false	true	true
HSEM global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



6. Software Pack Report

6.1. Software Pack selected

Vendor	Name	Version	Component
STMicroelectronics	X-CUBE-TOUCHGFX	4.25.0	Class : Graphics Group : Application Variant : TouchGFX Generator Version : 4.25.0

7. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32wb_bsdl.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32wb_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32wb_svd.zip
Board Manufacturing Specifications	https://www.st.com/resource/en/board_manufacturing_specification/stm32wb55cg_mb1487_ref_board.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval-tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers_stm32wbxm_wireless-modules_product_overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-entry-level-graphics.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-graphics-solution-overview.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-graphics-solutions-detailed.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wb.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wbvl.pdf

Flyers	https://www.st.com/resource/en/flyer/flstm32matter.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32wbxm.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32zigbee.pdf
White Papers	https://www.st.com/resource/en/white_paper/seamless-smart-home-connectivity-with-matter-whitepaper.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/stm32wb-rf-certificates.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/ble-thread-ftd-dynamic-thread-device-interoperability-certificate.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/full-thread-device-interoperability-certification.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/minimal-thread-device-interoperability-certification.pdf
Security Advisory	https://www.st.com/resource/en/security_advisory/sa0024-potential-isolation-issue-between-cpu1-and-cpu2-on-stm32wb5x-stm32wb3x-stm32wb1x-and-stm32wl5x-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2606-stm32-microcontroller-system-memory-boot-mode-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-

	applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5027-interfacing-pdm-digital-microphones-using-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5071-stm32wb-series-microcontrollers-ultralowpower-features-overview-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5105-getting-started-with-touch-sensing-control-on-stm32-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5155-stm32cube-mcu-package-examples-for-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5247-overtheair-application-and-wireless-firmware-update-for-stm32wb-series-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5378-stm32wb-series-microcontrollers-bringup-procedure-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5379-examples-of-at-commands-on-stm32wb-series-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5395-stm32wb-series-mcus-with-an-external-power-amplifier-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5434-onboard-antennas-reference-design-for-the-stm32wb-series-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5491-creating-manufacture-specific-clusters-on-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5492-persistent-data-management-zigbee-and-nonvolatile-memory-in-stm32wb-series-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5498-how-to-use-zigbee-clusters-templates-on-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5500-zsdk-api-implementation-for-zigbee-on-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5506-getting-started-with-zigbee-on-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5604-stm32wb-series-ble-interoperability-report-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5609-developing-zigbee-smart-energy-applications-on-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5613-getting-started-with-dynamicconcurrent-mode-ble--zigbee-on-stm32wb-series-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5627-stm32wb-series-zigbee-commissioning-guide-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5707-st-bluetooth-mesh-sensor-model-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5745-st-bluetooth-mesh-light-lc-server-model-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4760-quadspi-interface-on-stm32-microcontrollers-and-microprocessors--stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5732-developing-zigbee-sleepy-end-devices-on-stm32wb-series-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4899-stm32-microcontroller-gpio-hardware-settings-and-lowpower-consumption-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5612-esd-protection-of-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5292-how-to-build-a-bluetooth-low-energy-mesh-application-for-stm32wb-series-

	microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4991-how-to-wake-up-an-stm32-microcontroller-from-lowpower-mode-with-the-usart-or-the-lpuart-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4838-introduction-to-memory-protection-unit-management-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5165-how-to-develop-rf-hardware-using-stm32wb-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5290-getting-started-with-stm32wb-mcu-hardware-development-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4879-introduction-to-usb-hardware-and-pcb-guidelines-using-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5042-how-to-calibrate-the-hse-clock-for-rf-applications-on-stm32-wireless-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5246-how-to-use-smps-to-improve-power-efficiency-on-stm32wb-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5225-introduction-to-usb-typec-power-delivery-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5451-migrating-from-stm32wb3x5x-to-stm32wb3x5x-mcus-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5805-migrating-from-stm32wb1x5x-to-stm32wb1x5x-mcus-microcontrollers-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4894-how-to-use-eeprom-emulation-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5289-how-to-build-wireless-applications-with-stm32wb-mcus-stmicroelectronics.pdf

Application Notes	https://www.st.com/resource/en/application_note/an5886-guidelines-for-design-and-board-assembly-of-land-grid-array-packages-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5537-how-to-use-adc-oversampling-techniques-to-improve-signaltonoise-ratio-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5036-guidelines-for-thermal-management-on-stm32-applications-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5690-how-to-use-vrefbuf-peripheral-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4230-introduction-to-random-number-generation-validation-using-the-nist-statistical-test-suite-for-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2867-guidelines-for-oscillator-design-on-stm8afals-and-stm32-mcusmpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3236-how-to-increase-the-number-of-touchkeys-for-touch-sensing-applications-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an3960-guidelines-for-esd-for-touch-sensing-applications-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4013-introduction-to-timers-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4277-how-to-use-pwm-shutdown-for-motor-control-and-digital-power-conversion-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4299-how-to-improve-conducted-noise-robustness-for-touch-sensing-applications-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4310-how-to-choose-the-sampling-capacitor-for-touch-sensing-applications-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4312-how-to-design-surface-sensors-for-touch-sensing-applications-on-stm32-mcus-

	stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4316-how-to-tune-touch-sensing-applications-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4759-introduction-to-using-the-hardware-realtime-clock-rtc-and-the-tamper-management-unit-tamp-with-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4908-getting-started-with-usart-automatic-baud-rater-detection-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5156-introduction-to-security-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5224-introduction-to-dmamux-for-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5129-guidelines-for-meander-design-using-lowcost-pcb-antennae-with-24-ghz-radio-for-stm32wbwb0-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5543-guidelines-for-enhanced-spi-communication-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5185-how-to-use-stmicroelectronics-firmware-upgrade-services-for-stm32wb-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an5270-introduction-to-stm32wb-bluetooth-low-energy-wireless-interface-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/cd00211314-how-to-optimize-the-adc-accuracy-in-the-stm32-mcus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an2639-soldering-recommendations-and-package-information-for-leadfree-ecopack2-mcus-and-mpus-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an4435-guidelines-for-

for related Tools & Software	obtaining-ulcsaiec-607301603351-class-b-certification-in-any-stm32-application-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4657-stm32-inapplication-programming-iap-using-the-usart-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4841-digital-signal-processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5155-stm32cube-mcu-package-examples-for-stm32wb-series-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbpd-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an4865-lowpower-timer-lptim-applicative-use-cases-on-stm32-mcus-and-mpus-

& Software	stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4502-stm32-smbuspmbus-expansion-package-for-stm32cube-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5042-how-to-calibrate-the-hse-clock-for-rf-applications-on-stm32-wireless-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5952-how-to-use-cmake-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4635-how-to-optimize-lpuart-power-consumption-on-stm32-mcus-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5054-how-to-perform-secure-programming-using-stm32cubeprogrammer-stmicroelectronics.pdf
Errata Sheets	https://www.st.com/resource/en/errata_sheet/es0394-stm32wb55xxstm32wb35cx-device-errata-stmicroelectronics.pdf
Datasheet	https://www.st.com/resource/en/datasheet/dm00344191.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0214-stm32-cortexm4-mcus-and-mpus-programming-manual-stmicroelectronics.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0223-stm32-cortexm0-mcus-programming-manual-stmicroelectronics.pdf
Programming Manuals	https://www.st.com/resource/en/programming_manual/pm0271-guidelines-for-bluetooth-low-energy-stack-programming-on-stm32wbstm32wba-mcus-stmicroelectronics.pdf
Reference Manuals	https://www.st.com/resource/en/reference_manual/rm0434-multiprotocol-wireless-32bit-mcu-armbased-cortexm4-with-fpu-bluetooth-lowenergy-and-802154-radio-solution-stmicroelectronics.pdf
Technical Notes	https://www.st.com/resource/en/technical_note/tn1163-description-of-

& Articles	wlcsp-for-microcontrollers-and-recommendations-for-its-use-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1204-tape-and-reel-shipping-media-for-stm32-microcontrollers-in-bga-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1205-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-fpn-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1206-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-qfp-packages-stmicroelectronics.pdf
Technical Notes & Articles	https://www.st.com/resource/en/technical_note/tn1207-tape-and-reel-shipping-media-for-stm8-and-stm32-microcontrollers-in-so-packages-stmicroelectronics.pdf
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